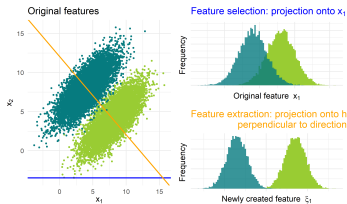
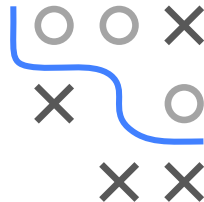


# Introduction to Machine Learning

## Feature Selection

## Feature Selection: Introduction



### Learning goals

- Too many features can be harmful in prediction
- Selection vs. extraction
- Types of selection methods

# INTRODUCTION

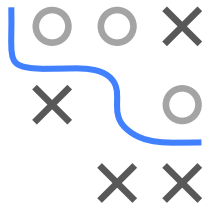
Feature selection:

Finding a well-performing, hopefully small set of features for a task.

Feature selection is critical for

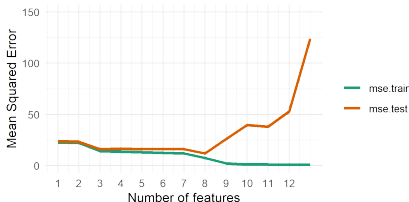
- reducing noise and overfitting
- improving performance/generalization
- enhancing interpretability by identifying most informative features

Features can be selected based on domain knowledge, or data-driven algorithmic approaches. We focus on the latter here.



# MOTIVATION

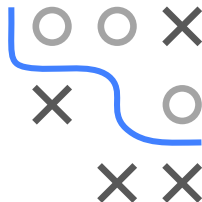
- Naive view:
  - More features  $\rightarrow$  more information  $\rightarrow$  discriminant power  $\uparrow$
  - Model is not harmed by irrelevant features since their parameters can simply be estimated as 0.
- In practice, irrelevant and redundant features can “confuse” learners (see **curse of dimensionality**) and worsen performance.
- Example: In linear regression,  $R^2$  is monotonically increasing in  $p$ , but adding irrelevant features leads to overfitting (capturing noise).



# SIZE OF DATASETS

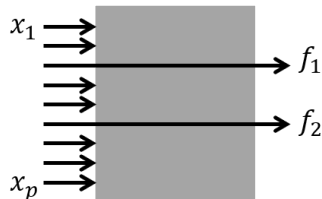
Many new forms of technical measurements and connected data leads to availability of extremely high-dimensional data sets.

- **Classical setting:** Up to around  $10^2$  features, feature selection might be relevant, but benefits often negligible.
- **Datasets of medium to high dimensionality:** At around  $10^2$  to  $10^3$  features, classical approaches can still work well, while principled feature selection helps in many cases.
- **High-dimensional data:**  $10^3$  to  $10^9$  or more features. Examples: micro-array / gene expression data and text categorization (bag-of-words features). If we also have few observations, scenario is called  $p \gg n$ .



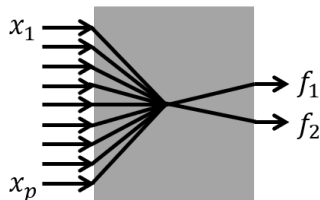
# FEATURE SELECTION VS. EXTRACTION

## Feature selection

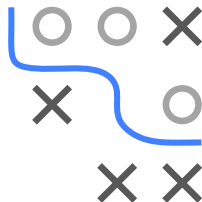


- Creates a subset of original features  $\mathbf{x}$  by selecting  $\tilde{p} < p$  features  $\mathbf{f}$ .
- Retains information on selected individual features.

## Feature extraction



- Maps  $p$  features in  $\mathbf{x}$  to  $\tilde{p}$  extracted features  $\mathbf{f}$ .
- Info on individual features can be lost through (non-)linear combination.



## TYPES OF FEATURE SELECTION METHODS

In rest of the chapter, we introduce different types of methods for FS:

**Example: embedded method (Lasso)** regularizing model params with  $L_1$  penalty enables “automatic” feature selection:

